Jin Huang (Hugo) PhD Candidate @ University of Edinburgh

Website: hugohuang.com **LinkedIn:** HugoHuang123 **GitHub:** Trenza1ore **Email:** hugo@hugohuang.com
If any information seem outdated, latest version of this resume is available at hugohuang.com
ORCID **Google Scholar** **OpenReview**

Education

PhD	**Artificial Intelligence and its Applications (AIAI)**	University of Edinburgh	**Since Oct. 2026**
Master	**MSc Artificial Intelligence**	University of Edinburgh	Sep. 2023 – Aug. 2024
Bachelor	**BSc Computer Science**	Cardiff University	Sep. 2020 – Jun. 2023

Current Position

Head of Developer & Open Source Maintainer @ Farama Foundation **Jun. 2026 – Current**

- Core team member, direct maintainer of HighwayEnv and Shimmy; also help to maintain other Farama projects such as Gymnasium (OpenAI Gym) and PettingZoo (basically multi-agent Gym) as the head of developer.
- We maintain many of the world's open-source Reinforcement Learning tools: 870+ GitHub contributors, 470M+ installations, and 36k+ public repositories on GitHub depend on us :-)

Previous Positions

Research Engineer @ Huawei – 2012 Lab / Retrieval & Agent Intelligence **Nov. 2024 – Sep. 2026**

- **2025 Annual Performance Review: Excellent** (top 10%).
- **Program Committee Member**, PromptEng workshop – co-located with the ACM Web Conference 2026.
- **Challenge Designer**, CCF Big Data & Computing Intelligence Contest 2026, with 1300+ contestants for my challenge.
- Received **Star of Edinburgh Award** for my contribution to Huawei's Agent System, openJiuwen. Top contributor, main code reviewer of Edinburgh, led development for Knowledge Base Retrieval, Graph Memory, Code Graph, etc.
- Helped to establish internal collaboration projects with other teams within Edinburgh Research Center & with teams from China HQ. Also was admin to most servers within RAI, maintaining internal model / database services.
- Helped with **openJiuwen-deepsearch**, claimed the BrowseComp-Plus leaderboard in 2025 with GPT-5-mini.
- Collaborated with programming language department on code conversion project to aid Huawei customers convert iOS or Android application code base to work on HarmonyOS, building code graph to reduce noise in agentic exploration.
- Led development for multiple close-source agent features, deployed to Huawei Cloud and Top-4 Banks in China.

Publications

2023 **BEV@DC: Bird's-Eye View Assisted Training for Depth Completion**
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023)
Wending Zhou, Xu Yan, Yinghong Liao, Yuankai Lin, Jin Huang, Gangming Zhao, Shuguang Cui, Zhen Li
Computer Vision (Depth Completion)

2025 **LLM Shots: Best Fired at System or User Prompts?**
WWW '25: Companion Proceedings of the ACM on Web Conference 2025
Umut Halil, Jin Huang, Damien Graux, Jeff Z. Pan
Natural Language Processing (In-Context Learning, Prompt Engineering)

2026 **Gnosis Deepsearch: Structured Search Space Navigation for Complex QA**
Findings of the Association for Computational Linguistics: EMNLP 2026
Umut Halil*, Le Zhang*, Jin Huang*, Yuwei Zhang*, Xinyu Hou*, Andre Melo, Shriram Piramanayagam, Amela Fejza, Ruofei Lai, Yang Ren, Pavlos Vougiouklis, Jeff Z. Pan
* co-first authors
Natural Language Processing (Deep Research, Multi-Agent)

Open Source Contributions Outside of Work

packaging.python.org Python's official packaging guide.
Contributor *Wrote sections regarding self-referential extras and best practice regarding combined extras.*

pip The package installer for Python.
Contributor *Added documentation & regression tests, marked official support for self-referential extras.*

Stable Baselines3
(26M downloads on PyPI)
RL Framework
Contributor

Stable Baselines3 (SB3) is a set of reliable implementations of Reinforcement Learning algorithms in PyTorch. It is the next major version of Stable Baselines. One of the most popular Reinforcement Learning frameworks.
Enhancement for rollout buffers, reduced memory usage (in SB3 & SBX - JAX version).

ViZDoom
(430k downloads on PyPI)
RL & CV Environment
Top Contributor

Cited in thousands of papers, ViZDoom allows developing AI that play Doom using only visual info. It powered precursors of world model like GameGAN and GameNGen!
Added build pipeline for Python binding's typing support, in-engine object categorization (Semantic Segmentation) and fixed various stuffs in GitHub Action & CMake, etc.

GitCode API
(62k downloads on PyPI)
Python SDK for GitCode
Author

An easy-to-use Python SDK / CLI tool / MCP Server (with MCPB support) for GitCode, second largest Chinese Git platform, built with user-centric API design. Since its creation, many projects from me and colleagues at Huawei have started to integrate it due to the ease of use, typing design, and documentation, now the most downloaded GitCode tool.

SB3 Extra Buffers
(12k downloads on PyPI)
RAM-Saving Tool for RL
Author

A collection of extra Stable Baselines3 buffer classes. Reducing memory usage drastically with minimal overhead via vectorized implementation of lossless compression algorithms. Experiments conducted with Atari games show $> 90\%$ memory save with negligible latency. Project's starred by maintainers of SB3 & OpenAI Gym, featured in SB3 documentation.

STD Raytracer
(Personal Project)
Ray-tracer / Path-tracer
Author

A ray-traced / path-traced renderer with support for custom models & textures, built from scratch with only C++ 17 standard library and nlohmann's json parser. Multi-threaded with bounding volume hierarchy acceleration, alternate frame rendering option, detailed logs and progress bar for user-friendliness. Support loading simple animation from Blender.

SegDoom
(Master's Thesis Project)
Play Doom with RL & CV
Author

Enhancing Reinforcement Learning in 3D Environments through Semantic Segmentation: A Case Study in ViZDoom explored the potential of applying Semantic Segmentation (SS) to 3D video game environment and analyzed issues in previous literature. Experiments conducted in ViZDoom showed significant improvement with SS mask as augmentation to RGB input, and decrease in memory usage was observed if SS is used as a replacement to RGB (see sb3-extra-buffers). Novel visualization methods were developed to evaluate RL agents' suitability for data collection. Custom Semantic Segmentation dataset was created.

Past Internships

Backend Developer *SenseTime*	Designed a tree-based RAG system for DevOps manuals, it preserves the structured nature of source materials and made use of keywords & synonyms to further provide alignment. Outperform official RAG implementation in SenseNova (SenseTime's LLM) at the time.	Jul. 2023 – Aug. 2023 Shenzhen, China
Research Assistant *Cardiff University*	Developed a dataset creation tool in Unity Engine for synthesizing image sequences with mirror segmentation.	Jun. 2022 – Aug. 2022 Cardiff, UK

Past Voluntary Works

CS Tutor *SCIE (High School)*	Worked as Computer Science tutor for GCSE students in my high school, Shenzhen College of International Education. All students that entered with C/D grade improved to above B in IGCSE Exam.	Sep. 2018 – Jun. 2019 Shenzhen, China

Skills

General Workplace Skill	Communication, Teamwork, Translation (Between English and Chinese), Presentation
AI R&D Specific Skill	Bridging gaps between scientists' and engineers' mindsets during mettings & discussions
Field of Research	Natural Language Processing, Reinforcement Learning, Computer Vision
Programming Language	Python, C++, C#, Java, MATLAB, ACS, etc.
Scientific Computing	NumPy, Numba, PyTorch, Sci-kit Learn, Gensim, FAISS, etc.
Hosting Services/Demos	Docker, Podman, Flask, FastAPI, Gradio, ECharts, Vis.js, etc.
Vector/Graph Database	Milvus, Chroma, Qdrant, Nebula, Neo4j, LadybugDB, Kuzu, SQLite-Vec, etc.
Agent Development	Tool & Resources (MCP, OpenAI tool), SDK (LangChain/Graph, openJiuwen, AutoGen)
Open Source Projects	GitHub Actions, Sphinx Documentation (Read the Docs), PyPI, writing pyproject.toml.